Condition Number

Zhenhuan Sun

1 Matrix Norm

Recall that the ℓ_2 norm of a vector $\mathbf{x} \in \mathbb{R}^d$ is defined as

$$\|\mathbf{x}\|_2 \coloneqq \sqrt{x_1^2 + \cdots + x_d^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

Similar to vectors, matrices also have norms. One commonly used matrix norm is the 2-norm, also called the spectral norm, which is defined as

$$\|\mathbf{A}\|_2 \coloneqq \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|_2. \tag{1}$$

The second equality follows from the fact that every nonzero $\mathbf{x}$ can be normalized to a unit vector $\mathbf{u} \coloneqq \mathbf{x}/\|\mathbf{x}\|_2$ and

$$\frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \frac{\|\mathbf{A}(\|\mathbf{x}\|_2 \mathbf{u})\|_2}{\|\mathbf{x}\|_2} = \|\mathbf{A}\mathbf{u}\|_2.$$

Therefore, $\|\mathbf{A}\|_2$ is the maximum length that can be obtained by applying $\mathbf{A}$ to a unit vector. If $\|\mathbf{A}\|_2 = k$ where $k > 1$, then $\mathbf{A}$ can stretch a unit vector to, at most, length k. If $\|\mathbf{A}\|_2 = k$ where $0 < k < 1$, then $\mathbf{A}$ shrinks every unit vector, and even in the least-shrunk direction, the resulting vector has length $k < 1$.

It can be shown that the matrix 2-norm is related to the singular value of the matrix. Squaring both sides of (1) gives

$$\begin{aligned}
\|\mathbf{A}\|_2^2 &= \max_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|_2^2 \\
&= \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x} \\
&= \max_{\|\mathbf{x}\|=1} \mathbf{x}^\top \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top \mathbf{x}.
\end{aligned}$$

where the last equality is obtained by performing eigendecomposition to the real symmetric matrix $\mathbf{A}^\top \mathbf{A}$. For any unit vector $\mathbf{x}$, define unit vector $\mathbf{z} \coloneqq \mathbf{Q}^\top \mathbf{x}$, then

$$\mathbf{x}^\top \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top \mathbf{x} = \mathbf{z}^\top \mathbf{\Lambda}\mathbf{z} = \sum_{i=1}^{d} \lambda_i z_i^2.$$

Since $z_i^2 \geq 0$ and $\sum_{i=1}^d z_i^2 = 1$, we have

$$\sum_{i=1}^d \lambda_i z_i^2 \leq \sum_{i=1}^d \lambda_{\max}(\mathbf{A}^\top \mathbf{A}) z_i^2 = \lambda_{\max}(\mathbf{A}^\top \mathbf{A}) \sum_{i=1}^d z_i^2 = \lambda_{\max}(\mathbf{A}^\top \mathbf{A}),$$

where the equality is achieved when $\mathbf{x}$ is the unit eigenvector associated with $\lambda_{\max}(\mathbf{A}^\top \mathbf{A})$. Recall that the singular values of $\mathbf{A}$ are defined as

$$\sigma_i(\mathbf{A}) = \sqrt{\lambda_i \left(\mathbf{A}^\top \mathbf{A}\right)}.$$

Combining the two results above gives

$$\|\mathbf{A}\|_2 = \sigma_{\max}(\mathbf{A}). \tag{2}$$

2 Condition Number

For an invertible matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, its condition number is defined, using the matrix 2-norm, as

$$\kappa(\mathbf{A}) \coloneqq \|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2. \tag{3}$$

Using the matrix norm property $\|\mathbf{AB}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{B}\|_2$, it can be readily seen that $\kappa(\mathbf{A}) \geq 1$.

Denote the singular value decomposition of $\mathbf{A}$ as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top,$$

where $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d \times d}$ are orthogonal matrices and

$$\mathbf{\Sigma} = \operatorname{diag}\left(\sigma_1(\mathbf{A}), \sigma_2(\mathbf{A}), \ldots, \sigma_d(\mathbf{A})\right),$$

with $\sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_d(\mathbf{A})$. Since $\mathbf{A}$ is invertible, it has full rank, by the rank-nullity theorem it has a trivial null space. If any singular value were zero, the corresponding right singular vector $\mathbf{v}_i \neq 0$ would satisfy $\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i = 0$, giving a nonzero vector in the null space. Therefore, all singular values of $\mathbf{A}$ must be strictly positive, that is $\sigma_d(\mathbf{A}) > 0$. In addition, the invertibility of $\mathbf{A}$ also suggests that

$$\mathbf{A}^{-1} = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^\top,$$

where

$$\mathbf{\Sigma}^{-1} = \operatorname{diag}\left(\frac{1}{\sigma_1(\mathbf{A})}, \frac{1}{\sigma_2(\mathbf{A})}, \ldots, \frac{1}{\sigma_d(\mathbf{A})}\right).$$

Therefore, the largest singular value of $\mathbf{A}^{-1}$ is the reciprocal of the smallest singular value of $\mathbf{A}$. Using (2), we therefore obtain

$$\|\mathbf{A}^{-1}\|_2 = \sigma_{\max}(\mathbf{A}^{-1}) = \frac{1}{\sigma_{\min}(\mathbf{A})}.$$

Substituting this expression into (3) gives

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})}. \tag{4}$$

If $\mathbf{A}$ is symmetric, then $\mathbf{A}^\top\mathbf{A} = \mathbf{A}^2$. Let $\mathbf{q}_i$ and λ_i denote an eigenvector and the corresponding eigenvalue of $\mathbf{A}$, respectively. Then

$$(\mathbf{A}^\top\mathbf{A})\mathbf{q}_i = \mathbf{A}(\mathbf{A}\mathbf{q}_i) = \mathbf{A}(\lambda_i\mathbf{q}_i) = \lambda_i^2\mathbf{q}_i.$$

Therefore, $\mathbf{q}_i$ is also an eigenvector of $\mathbf{A}^\top\mathbf{A}$, with corresponding eigenvalue λ_i^2. Thus, $\mathbf{A}$ and $\mathbf{A}^\top\mathbf{A}$ share the same eigenvectors, while the eigenvalues of $\mathbf{A}^\top\mathbf{A}$ are the squares of those of $\mathbf{A}$. If $\mathbf{A}$ is also positive definite, then all eigenvalues of $\mathbf{A}$ are strictly positive, by definition of singular values, we have

$$\sigma_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}^\top\mathbf{A})} = \sqrt{\lambda_i^2} = |\lambda_i| = \lambda_i.$$

It then follows from (4) that

$$\kappa(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})}. \tag{5}$$

Therefore, for a symmetric positive definite matrix, the condition number is the ratio between its largest and smallest eigenvalues. A matrix with a large condition number, i.e., $\kappa(\mathbf{A}) \gg 1$, is called *ill-conditioned*, while a matrix with a condition number close to 1 is called *well-conditioned*.

Consider the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with an invertible matrix $\mathbf{A}$. If $\mathbf{b}$ is perturbed by a small amount $\Delta\mathbf{b}$, then the solution of the linear system also changes by some $\Delta\mathbf{x}$, that is

$$\mathbf{A}(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{b} + \Delta\mathbf{b}.$$

Since $\mathbf{A}\mathbf{x} = \mathbf{b}$, subtracting $\mathbf{b}$ from both sides and multiply both sides by $\mathbf{A}^{-1}$ gives $\Delta\mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b}$. Taking norms on both sides and using the matrix norm property $\|\mathbf{A}\mathbf{x}\|_2 \leq \|\mathbf{A}\|_2\|\mathbf{x}\|_2$ gives

$$\|\Delta\mathbf{x}\|_2 \leq \|\mathbf{A}^{-1}\|_2\|\Delta\mathbf{b}\|_2.$$

In addition, taking norms on both sides of $\mathbf{A}\mathbf{x} = \mathbf{b}$ and using the same property gives $\|\mathbf{b}\|_2 = \|\mathbf{A}\mathbf{x}\|_2 \leq \|\mathbf{A}\|_2\|\mathbf{x}\|_2$, which implies

$$\frac{1}{\|\mathbf{x}\|_2} \leq \frac{\|\mathbf{A}\|_2}{\|\mathbf{b}\|_2}.$$

Combining the above two inequalities, we have

$$\begin{aligned}\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}\|_2} &\leq \|\mathbf{A}^{-1}\|_2\|\Delta\mathbf{b}\|_2\frac{\|\mathbf{A}\|_2}{\|\mathbf{b}\|_2} \\ &= \kappa(\mathbf{A})\frac{\|\Delta\mathbf{b}\|_2}{\|\mathbf{b}\|_2}.\end{aligned}$$

Therefore, the condition number bounds how much a relative change in $\mathbf{b}$ can be amplified into a relative change in the solution $\mathbf{x}$. The larger the condition number, the larger the worst case amplification. Roughly speaking, condition numbers provide a measure of how sensitive the solution of a linear system $\mathbf{Ax} = \mathbf{b}$ is to changes in $\mathbf{b}$. The larger the condition number, the more sensitive the system is to changes in $\mathbf{b}$. The smaller the condition number, the less sensitive the system is to changes in $\mathbf{b}$.

3 A Numeric Example

Consider the following two unregularized linear systems:

$$\mathbf{Ax} = \mathbf{b}_1, \tag{6}$$
$$\mathbf{Ax} = \mathbf{b}_2, \tag{7}$$

where

$$\mathbf{A} = \begin{bmatrix} 0.01 & 0.1 \\ 0.1 & 2 \end{bmatrix}, \qquad \mathbf{b}_1 = \begin{bmatrix} 0.1 \\ 2 \end{bmatrix}, \qquad \mathbf{b}_2 = \begin{bmatrix} 0.1 \\ 2.1 \end{bmatrix}.$$

The two linear systems (6) and (7) are almost identical. The same matrix $\mathbf{A}$ is used in both linear system, $\mathbf{b}_1$ and $\mathbf{b}_2$ differ by only 0.1 in their second entry. Since the two linear systems differ only slightly, we might naturally expect their solutions to be similar as well. However, if we solve them, we obtain

$$\mathbf{x}_1^\star = \mathbf{A}^{-1}\mathbf{b}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$
$$\mathbf{x}_2^\star = \mathbf{A}^{-1}\mathbf{b}_2 = \begin{bmatrix} -1 \\ 1.1 \end{bmatrix}.$$

If we interpret each $\mathbf{x}_i$ as a parameter vector, that is $\mathbf{x}_i \coloneqq \begin{bmatrix} w_i & b_i \end{bmatrix}^\top$, then the solutions to (6) and (7) correspond to the following two linear functions:

$$y = 0 \cdot x + 1$$
$$y = -x + 1.1.$$

Plotting these two linear functions gives Figure 1. The observation here is that, when the right hand side of the linear system $\mathbf{Ax} = \mathbf{b}$ changes slightly from $\mathbf{b}_1$ to $\mathbf{b}_2$, the corresponding solution changes substantially from $\mathbf{x}_1^\star$ to $\mathbf{x}_2^\star$.

Now consider the following two regularized linear systems:

$$(\mathbf{A} + \lambda\mathbf{I})\mathbf{x} = \mathbf{b}_1, \tag{8}$$
$$(\mathbf{A} + \lambda\mathbf{I})\mathbf{x} = \mathbf{b}_2, \tag{9}$$

where $\lambda = 0.05$, and

$$\mathbf{A} = \begin{bmatrix} 0.01 & 0.1 \\ 0.1 & 2 \end{bmatrix}, \qquad \mathbf{b}_1 = \begin{bmatrix} 0.1 \\ 2 \end{bmatrix}, \qquad \mathbf{b}_2 = \begin{bmatrix} 0.1 \\ 2.1 \end{bmatrix}.$$

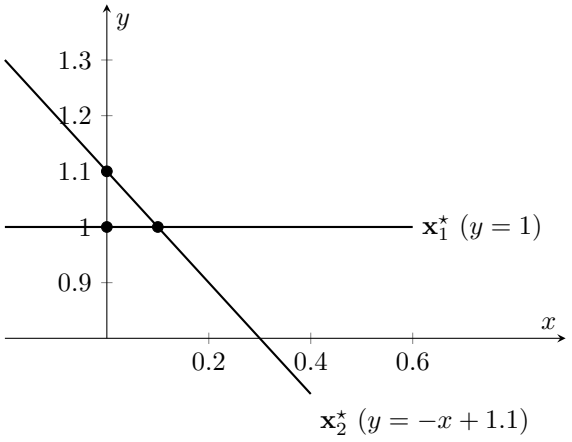


Figure 1: The linear functions correspond to the solutions of (6) and (7).

Since $\lambda = 0.05$, the matrix $\mathbf{A} + \lambda\mathbf{I}$ is

$$\mathbf{A} + \lambda\mathbf{I} = \begin{bmatrix} 0.06 & 0.1 \\ 0.1 & 2.05 \end{bmatrix}.$$

The two regularized linear systems (8) and (9) are again almost identical. They share the same matrix $\mathbf{A} + \lambda\mathbf{I}$, while $\mathbf{b}_1$ and $\mathbf{b}_2$ differ by only 0.1 in their second entry. Solving them gives

$$\mathbf{x}_3^\star = (\mathbf{A} + \lambda\mathbf{I})^{-1}\mathbf{b}_1 \approx \begin{bmatrix} 0.0442 \\ 0.9735 \end{bmatrix},$$
$$\mathbf{x}_4^\star = (\mathbf{A} + \lambda\mathbf{I})^{-1}\mathbf{b}_2 \approx \begin{bmatrix} -0.0442 \\ 1.0266 \end{bmatrix}.$$

Observe that these two solutions are much closer to each other than the solutions of the unregularized systems. If we again interpret each $\mathbf{x}_i$ as a parameter vector, then the solutions to (8) and (9) correspond to the following two linear functions:

$$y = 0.0442x + 0.9735,$$
$$y = -0.0442x + 1.0266.$$

Plotting these two linear functions gives Figure 2. The observation here is that, when the right hand side of the linear system $(\mathbf{A} + \lambda\mathbf{I})\mathbf{x} = \mathbf{b}$ changes slightly from $\mathbf{b}_1$ to $\mathbf{b}_2$, the corresponding solution changes only slightly from $\mathbf{x}_3^\star$ to $\mathbf{x}_4^\star$.

Therefore, by adding the regularization term $\lambda\mathbf{I}$ to $\mathbf{A}$, we observe that the solution of the regularized linear system becomes less sensitive to small changes in $\mathbf{b}$ compared to the solution of the unregularized linear system. The underlying reason is that adding the regularization term reduces the condition number of

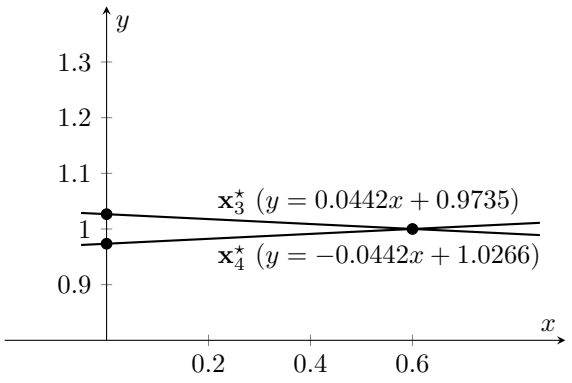


Figure 2: The linear functions corresponding to the solutions of (8) and (9).

the matrix $\mathbf{A}$. Using (5), it can be calculated that the condition number of $\mathbf{A}$ and $\mathbf{A} + \lambda\mathbf{I}$ are

$$\kappa(\mathbf{A}) \approx 402,$$
$$\kappa(\mathbf{A} + \lambda\mathbf{I}) \approx 37.$$

Since $\mathbf{A}$ has a larger condition number than $\mathbf{A} + \lambda\mathbf{I}$, the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is more sensitive to changes in $\mathbf{b}$ than the solution of $(\mathbf{A} + \lambda\mathbf{I})\mathbf{x} = \mathbf{b}$.

The linear systems considered above arise naturally from the closed-form solutions of basis function regression. In unregularized and regularized basis function regression, the optimal weight vectors have the form

$$\mathbf{w}^\star = (\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{y},$$
$$\mathbf{w}^\star = (\mathbf{B}^\top\mathbf{B} + \lambda\mathbf{I})^{-1}\mathbf{B}^\top\mathbf{y},$$

which are obtained by solving the linear systems

$$\mathbf{B}^\top\mathbf{B}\mathbf{w} = \mathbf{B}^\top\mathbf{y}, \tag{10}$$
$$(\mathbf{B}^\top\mathbf{B} + \lambda\mathbf{I})\mathbf{w} = \mathbf{B}^\top\mathbf{y}, \tag{11}$$

from which we can identify

$$\mathbf{A} = \mathbf{B}^\top\mathbf{B}, \qquad \mathbf{b} = \mathbf{B}^\top\mathbf{y}.$$

Therefore, if there is noise in $\mathbf{y}$, then $\mathbf{b}$ will be slighted perturbed. As illustrated by the previous example, the solution of (10) can be highly sensitive to such perturbations when $\mathbf{B}^\top\mathbf{B}$ is ill-conditioned. Adding the regularization term $\lambda\mathbf{I}$ improves the conditioning of the linear system, making the solution of (11) less sensitive to noise in $\mathbf{y}$. Consequently, regularization reduces the influence of noise on the learned model parameters and helps alleviate overfitting.